stm32_devboard_routed

Design Report

kicad-tools 0.20.0

Rev 1 | 2026-09-10 | jlcpcb-tier1

Board Summary

Property	Value
Layers	2 copper (F.Cu, B.Cu)
Footprints	17 (15 SMD, 2 THT, 0 other)
Nets	12
Traces	158 segments
Vias	29
Board Size	60.0 x 40.0 mm

Design Overview

Theory of Operation

STM32F103C8 Development Board

End-to-end design example

Demonstrates circuit blocks API

Power Architecture

Power Rails: +5V, +3.3V, GND

Regulator	Device
U1	MCP1825S-3302E/DB

Assembly Notes

1 fine-pitch component; 1 polarized component

- **Fine-pitch components**: 1 (U2)
- **Polarized components**: 1 – check orientation markings

ERC Status

Native KiCad ERC: **0 errors, 0 warnings**. The real MCP1825S pinout and portable symbol library are checked; see native-erc.json.

Schematic Overview

Schematic: stm32_devboard

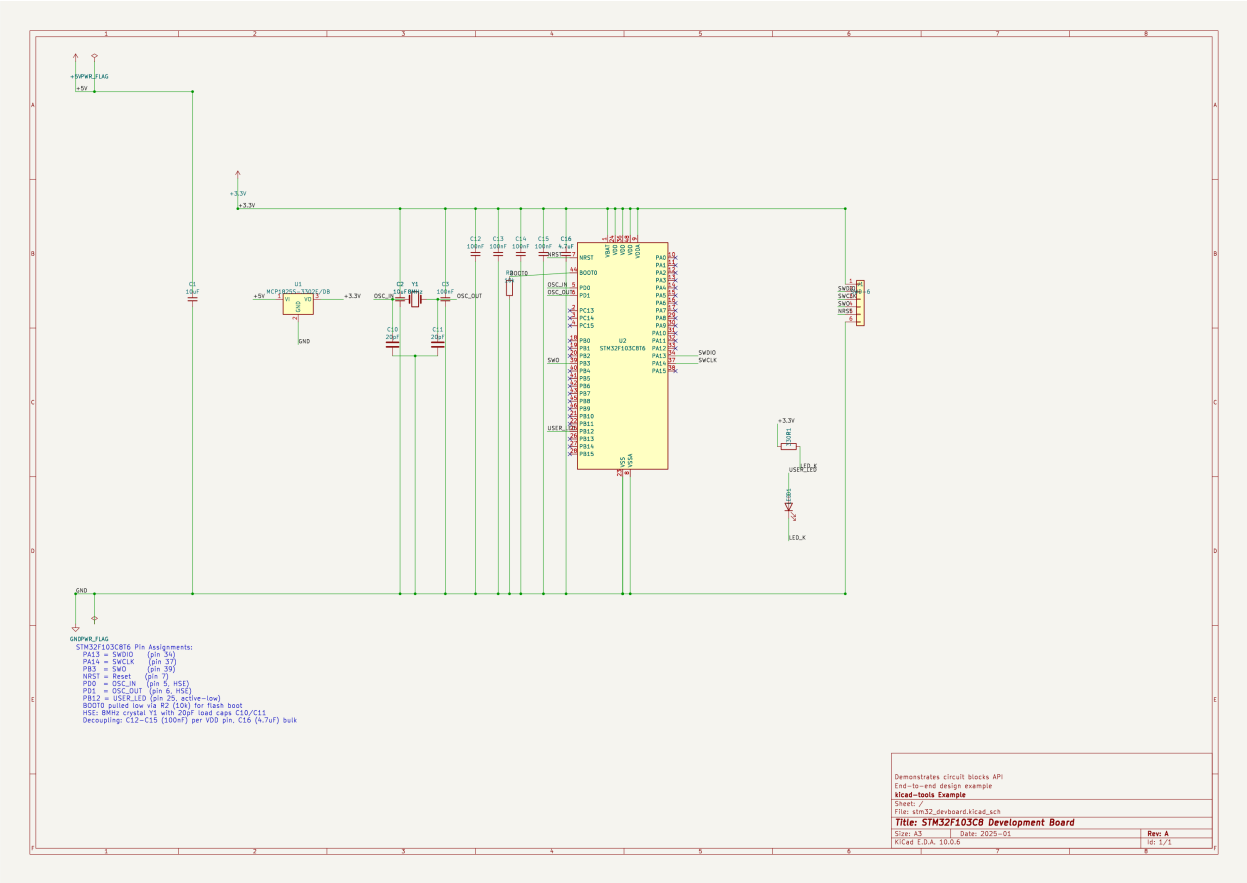


Figure 1: Schematic: stm32_devboard

PCB Layout

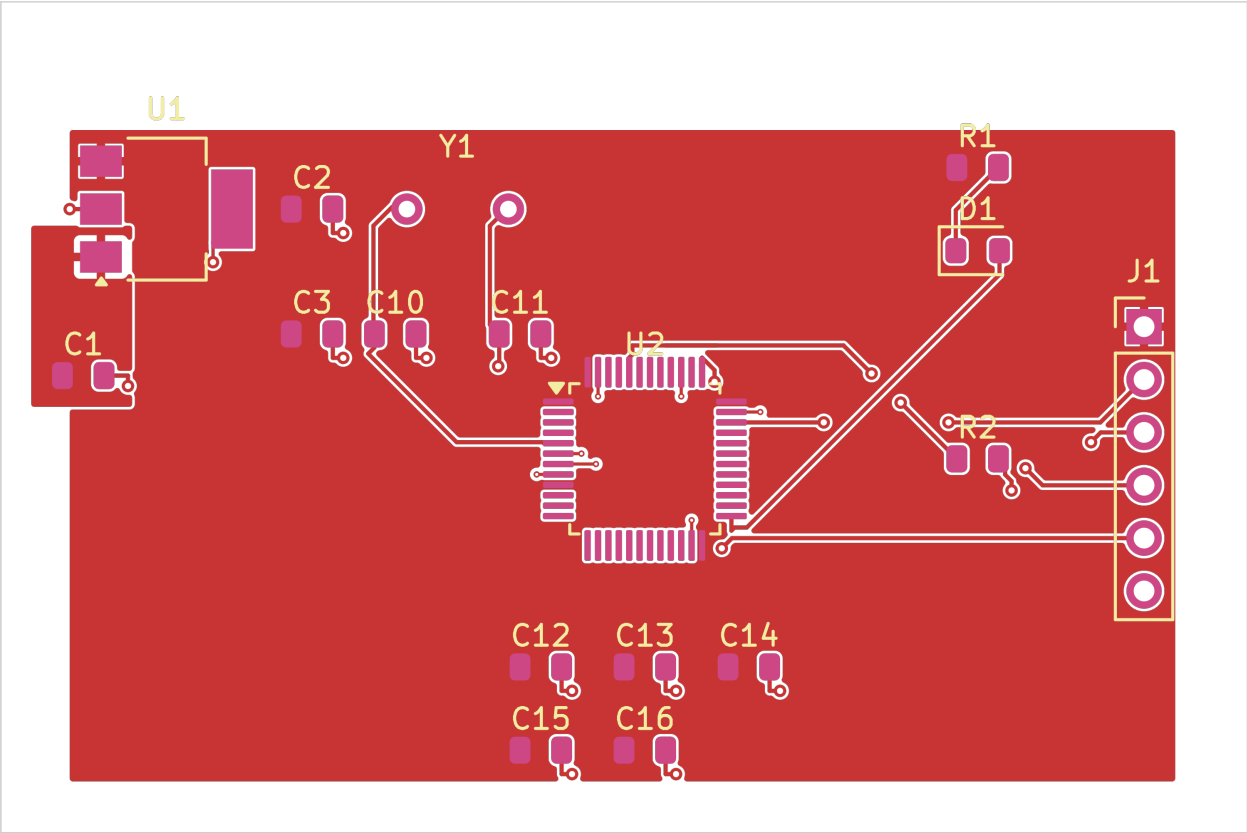


Figure 2: PCB Front

Copper

Assembly

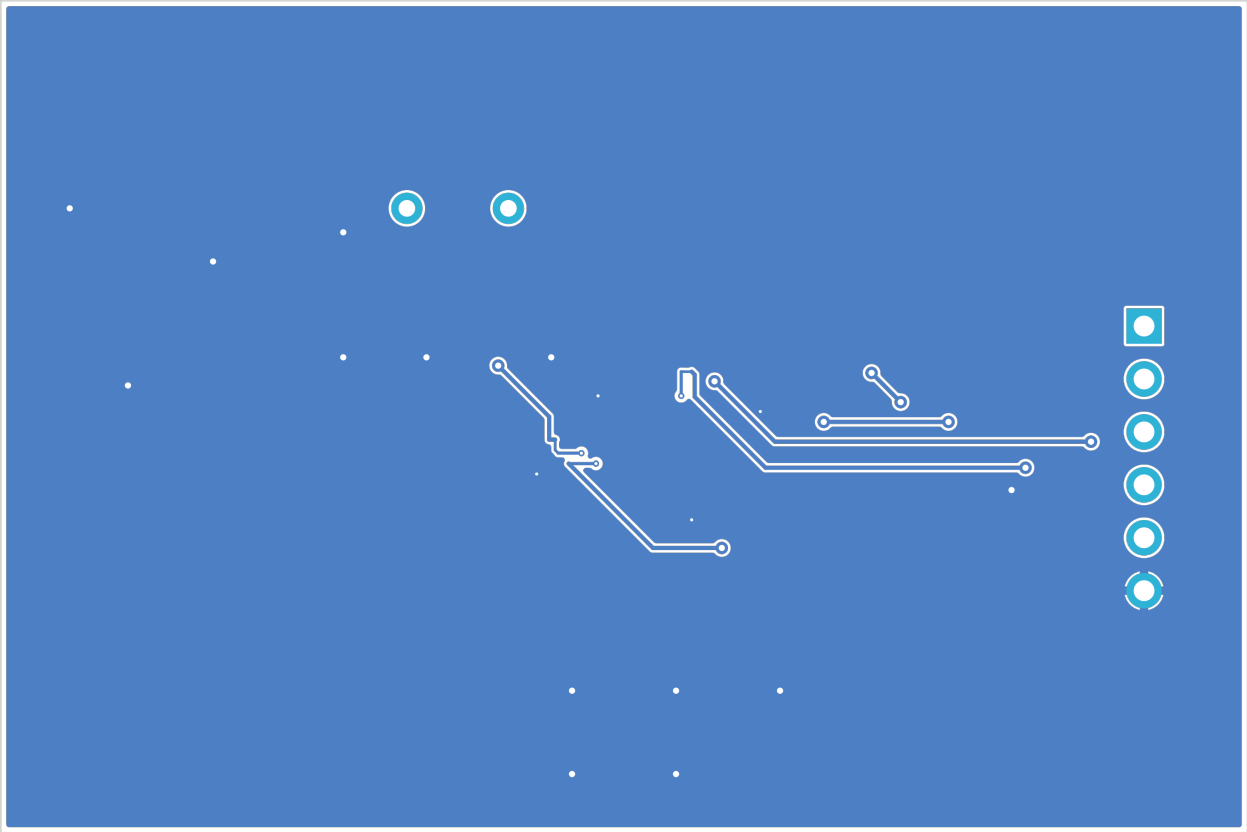


Figure 3: PCB Back

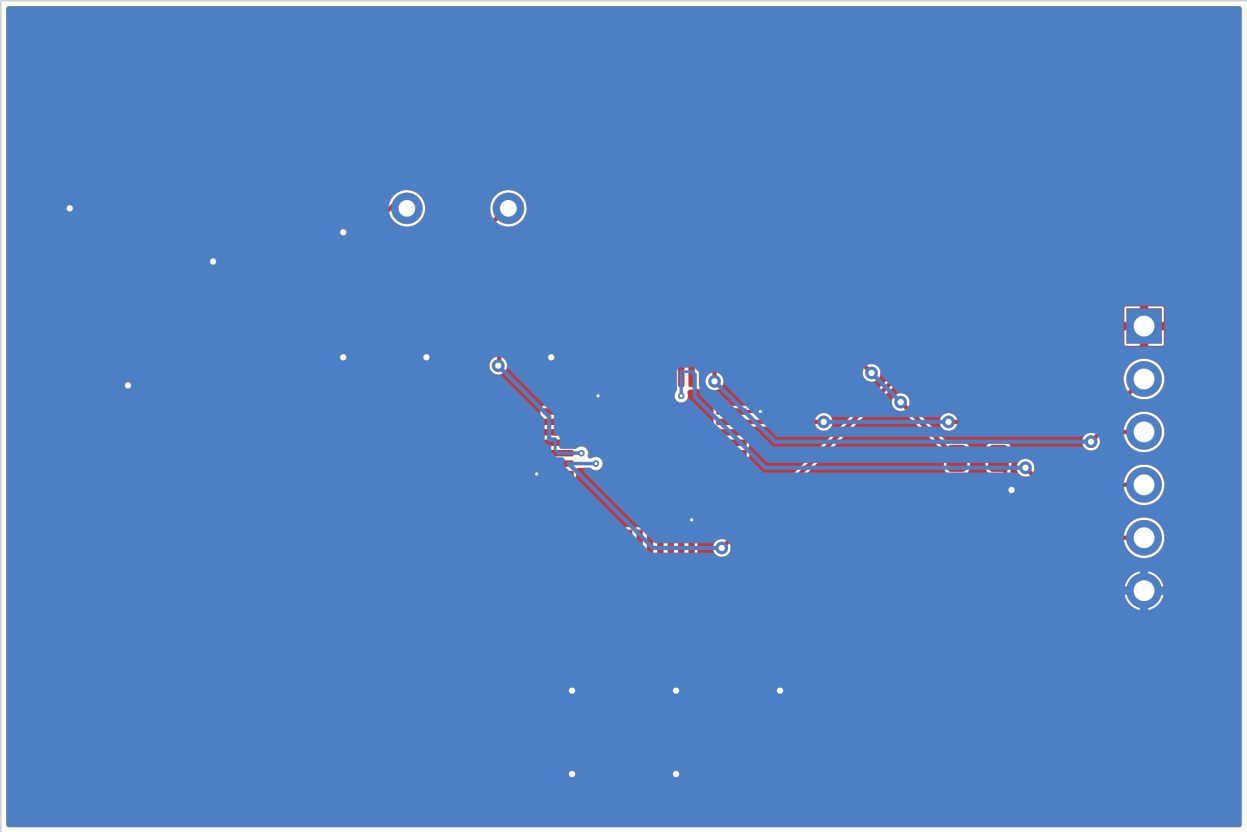


Figure 4: PCB Copper

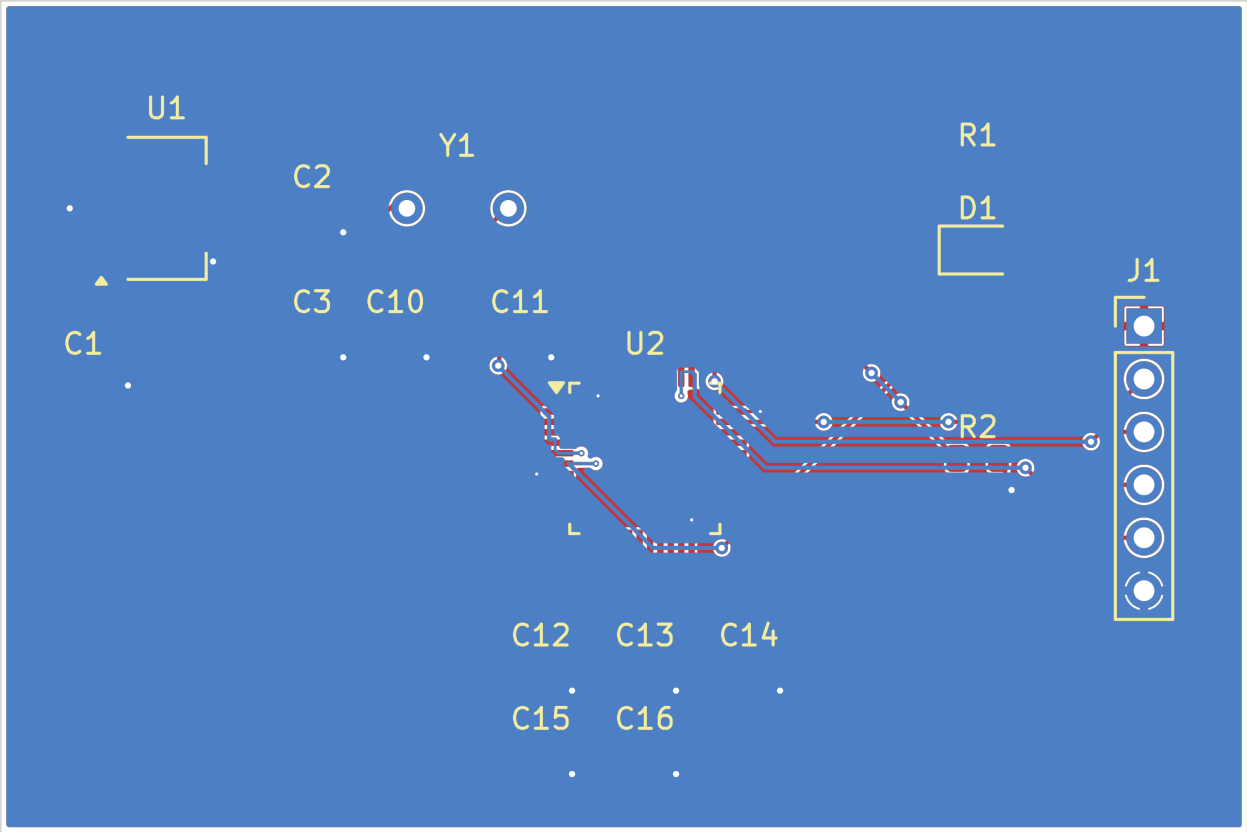


Figure 5: Assembly

Copper Layers

F.Cu

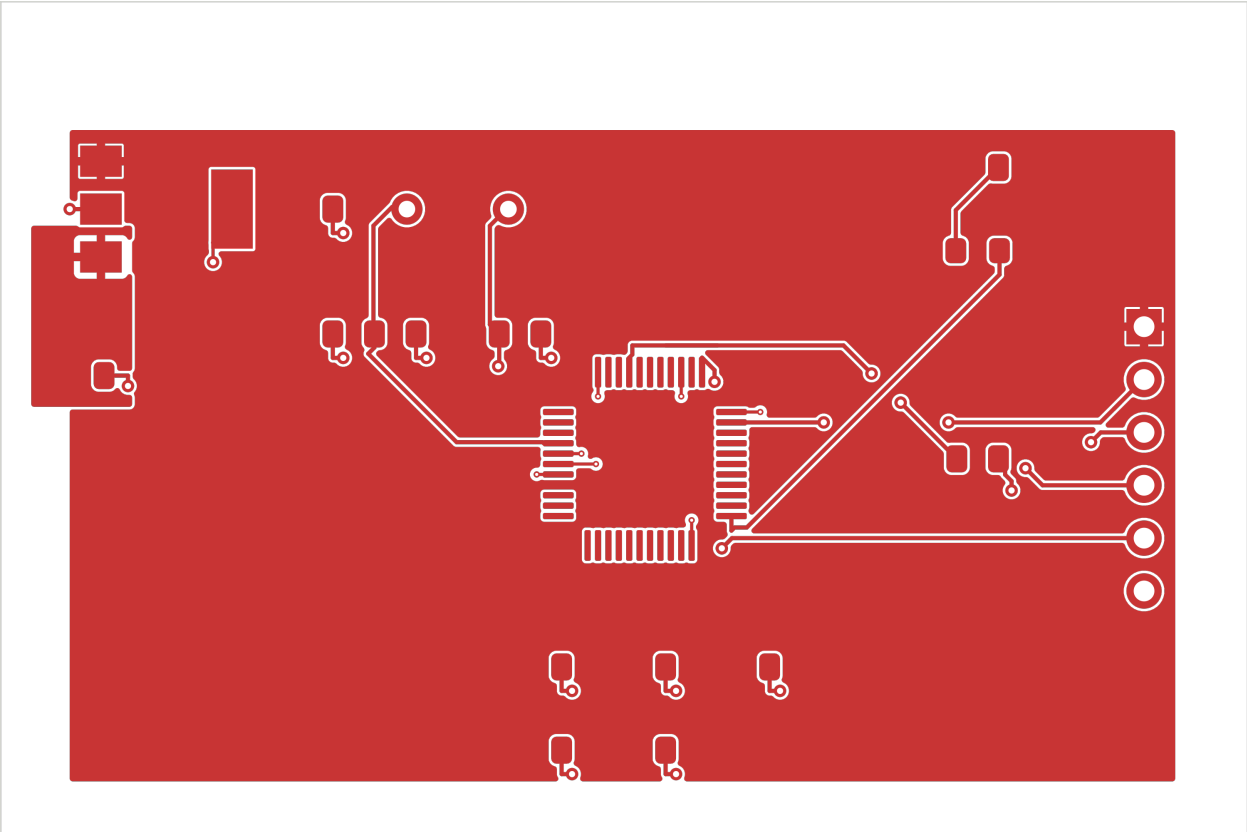


Figure 6: F.Cu

B.Cu

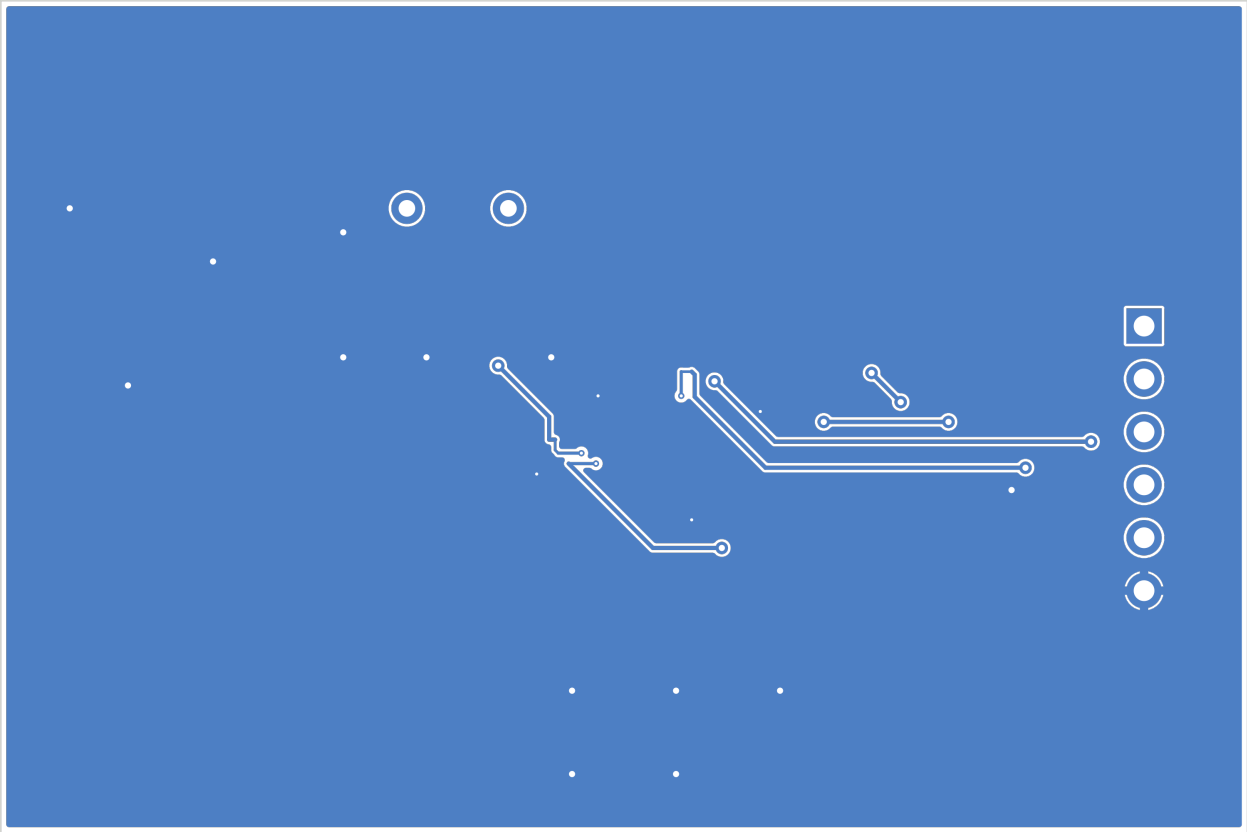


Figure 7: B.Cu

Bill of Materials

Value	Package	Qty	References	MPN	LCSC
100nF	C_0805_2012Metric	5	C3, C12, C13, C14, C15		
10uF	C_0805_2012Metric	2	C1, C2		
20pF	C_0805_2012Metric	2	C10, C11		
4.7uF	C_0805_2012Metric	1	C16		
LED	LED_0805_2012Metric	1	D1		
SWD-6	PinHeader_1x06_P2.54mm_Vertical	1	J1		
10k	R_0805_2012Metric	1	R2		
330R	R_0805_2012Metric	1	R1		
MCP1825S-3302E/DB	SOT-223-3_TabPin2	1	U1	MCP1825S-3302E/DB	C148031
STM32F103C8T6	LQFP-48_7x7mm_P0.5mm	1	U2		
8MHz	Crystal_HC49-4H_Vertical	1	Y1		

DRC Status

Metric	Count
Errors	0
Warnings	0

PASS: 0 errors, 0 warnings with the explicitly reviewed two-layer mechanical drilling process. Ordinary through vias use a minimum 0.15mm drill, 0.30mm diameter and 0.075mm annular ring. This paid JLC drilling option is documented in manufacturing-requirements.json. No findings are suppressed. All eight SMT-land drill overlaps were repaired with explicit connectivity tails.

Native DRC also has zero violations and opens. Schematic/copper LVS is clean. See check-report.json for complete checks and override provenance, native-drc.json for independent native evidence, and fill-consistency.json for exact exported-copper parity against an independent native refill.

Manufacturing Readiness

READY for fabrication and assembly with the specified process. Select the paid 0.15mm mechanical-drill option, two layers, 1.6mm FR4, 1oz copper and tented through vias. There are no laser microvias or drill overlaps with SMT lands. U1 must be MCP1825S-3302E/DB (C148031); do not fit AMS1117.

Use the 17-placement BOM/CPL: 15 SMT parts plus the through-hole J1 and Y1, which JLCPCB assembles (procurement must review those rows; see procurement-review.md). Power only from regulated 5V. First-article supply, thermal/load, SWD, crystal, reset and LED tests remain unperformed; see DESIGN_REVIEW.md. Geometric readiness does not claim physical qualification.

Routing Status

Metric	Value
Signal Net Completion	100.0% (9/9)
Overall Completion	91.7%
Complete Nets	11 / 12
Zone-Connected Nets	3
Incomplete Nets	1
Unconnected Pads	1

Zone-Connected Nets

- +3.3V
- +5V
- GND

Incomplete Nets

- GND

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~0.88 USD
Components (estimated)	~1.36 USD
Assembly (estimated)	~2.02 USD
Total (estimated)	**~4.26 USD**
Batch Quantity	5
Batch Total (estimated)	~21.3 USD